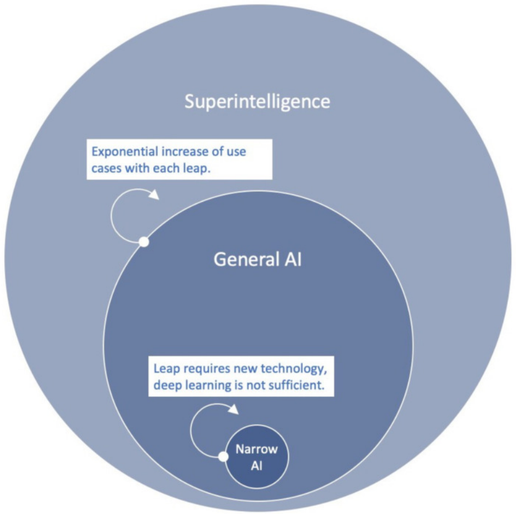
Inteligência Artificial e….

“FRAMEWORKS”(?)

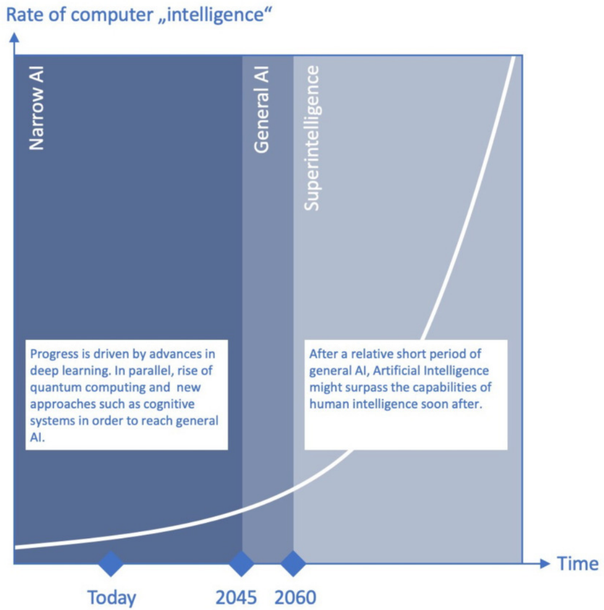
Inteligência Artificial e....

“QUADROS”(?)

Use Cases Increase Exponentially:

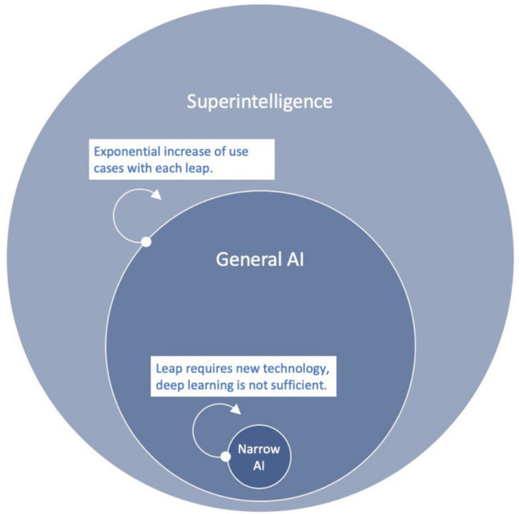


Accelerated Growth:

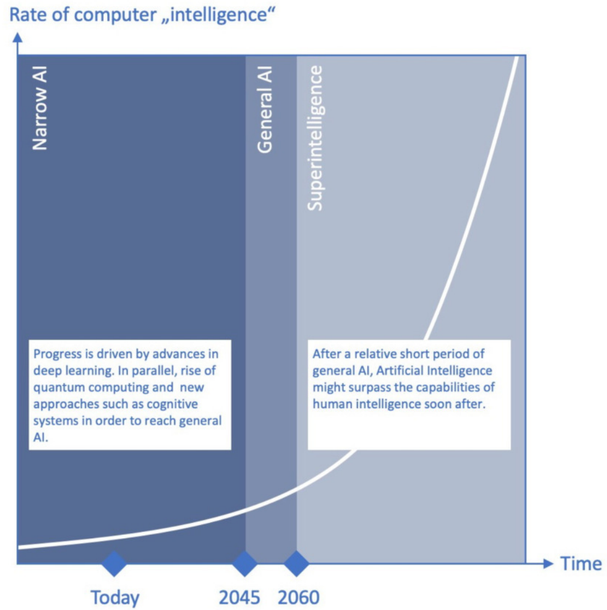


REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

Use Cases Increase Exponentially:



Accelerated Growth:

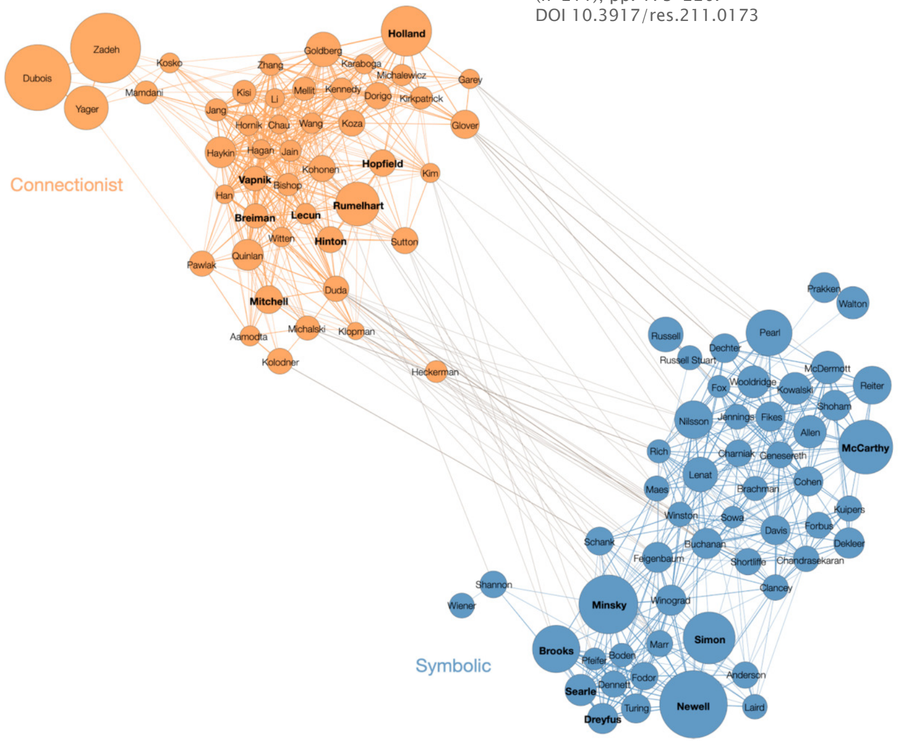


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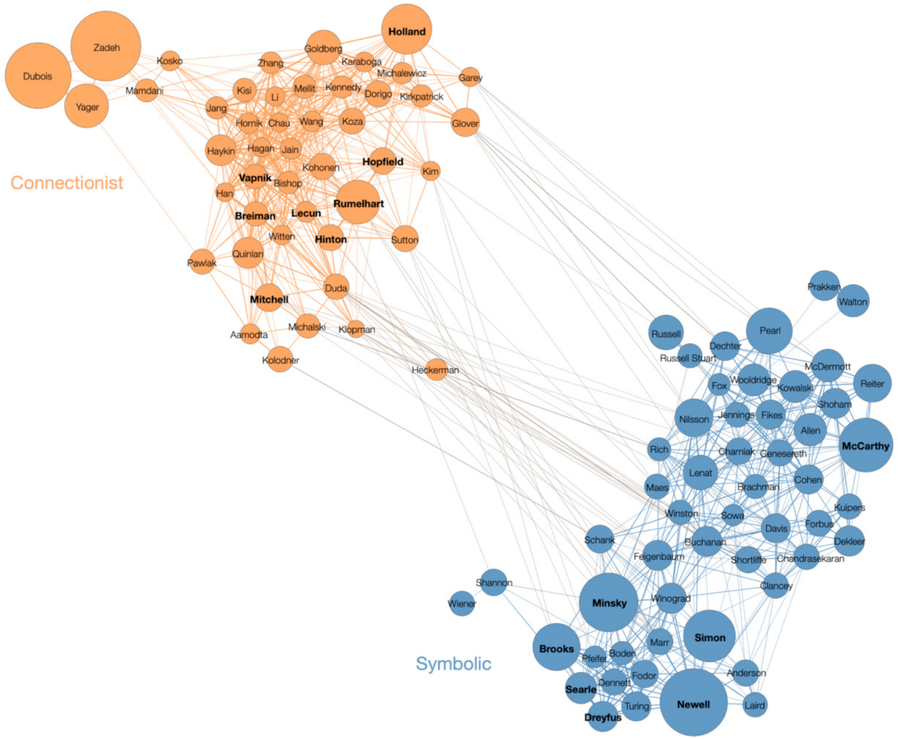
Co-citation network of the 100 most cited authors in scientific publications mentioning “Artificial Intelligence

REF: from
Dominique Cardon, Jean-Philippe Cointet et Antoine Mazières, «Neurons spike back. The
invention of inductive machines and the artificial intelligence controversy », Réseaux 2018/5
(n° 211), pp. 173-220.
DOI 10.3917/res.211.0173



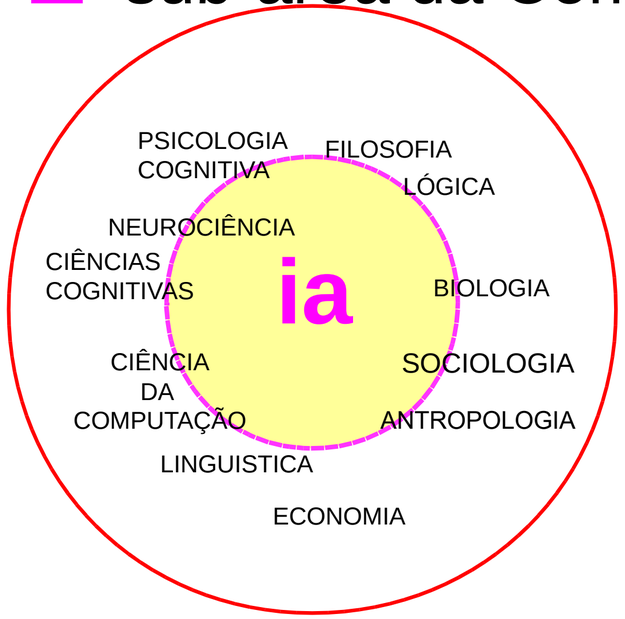
Rede de cocitação dos 100 autores mais citados em publicações científicas que mencionam “Inteligência Artificial”

REF: de Dominique Cardon, Jean-Philippe Cointet e Antoine Mazières, «Neurons spike back. A invenção das máquinas indutivas e a controvérsia da inteligência artificial », Réseaux 2018/5 (n° 211), pp. DOI 10.3917/res.211.0173



“Inteligência” Artificial(?)

NÃO É sub-área da Computação



“Inteligência” Artificial(?)

NÃO i subárea da Computação



IA - empresas



REF: https://www.topbots.com/essential-landscape-overview-enterprise-artificial-intelligence/

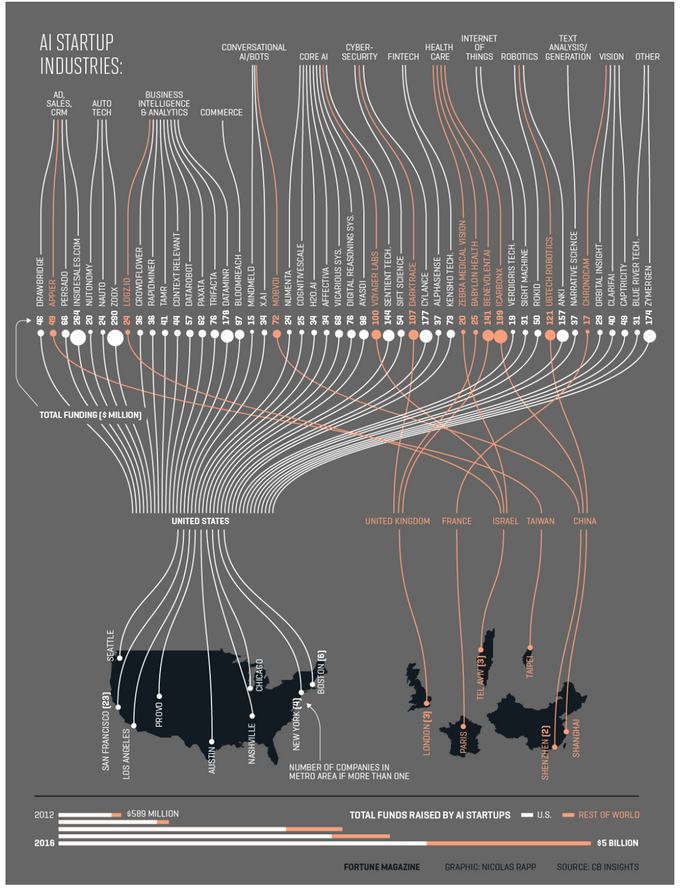
IA - empresas



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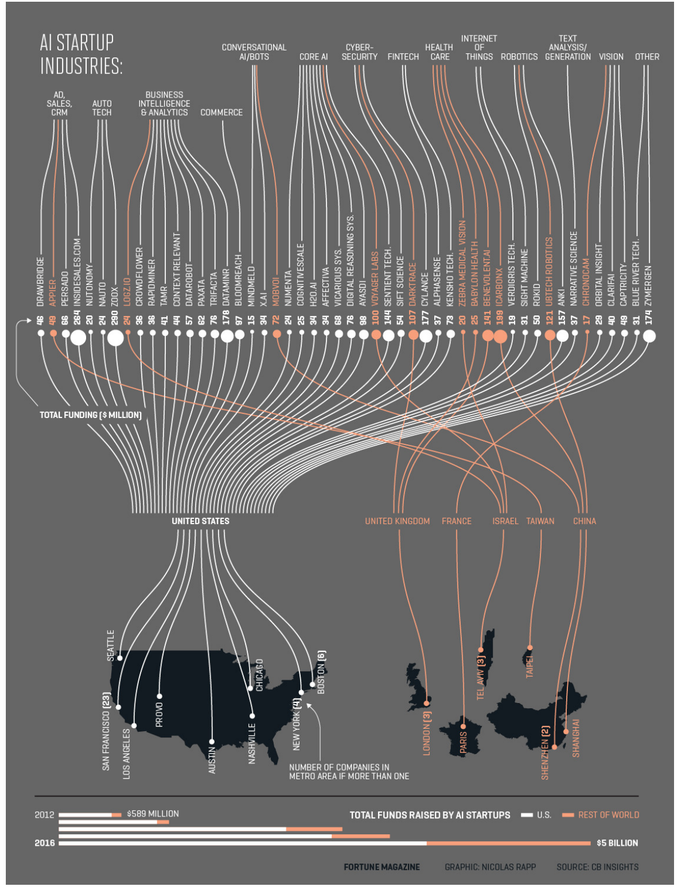
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IA – startups (como era - 2017)

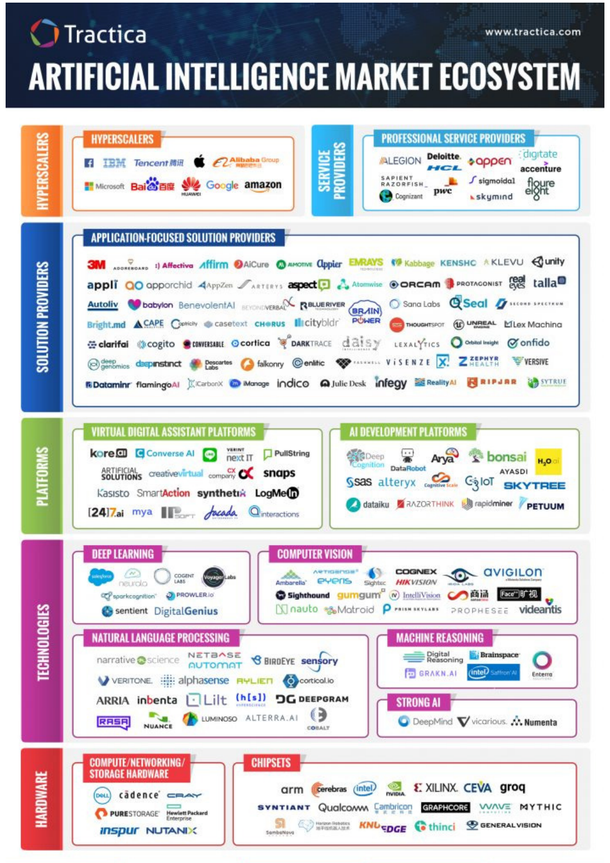


REF: https://content.fortune.com/wp-content/uploads/2017/02/rap_ai_web.png

IA – startups (era como - 2017)

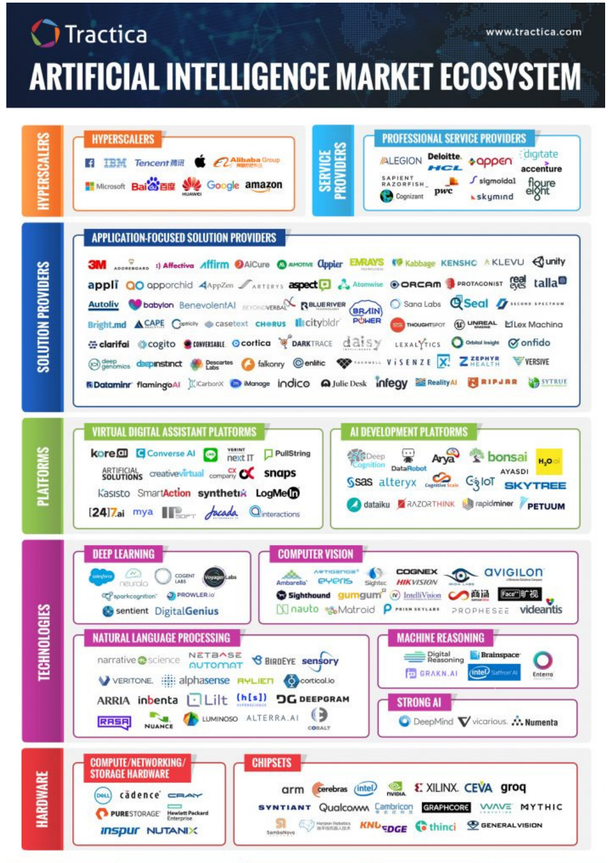


[6] REF: https://content.fortune.com/wp-content/uploads/2017/02/rap□ai□web.png



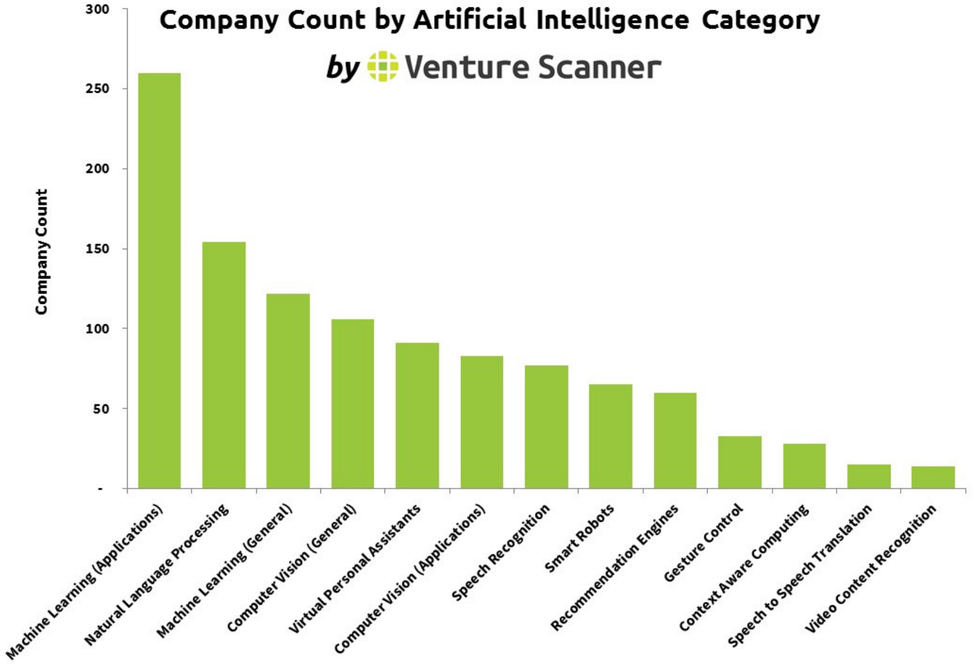
IA – ecosistema de empresas

REF: https://www.tractica.com/newsroom/press-releases /the-artificial-intelligence-market-ecosystem-is-Expanding-and-diversifying-rapidly-with-more-than-1000-companies-actively-driving-innovation-in-the-market/

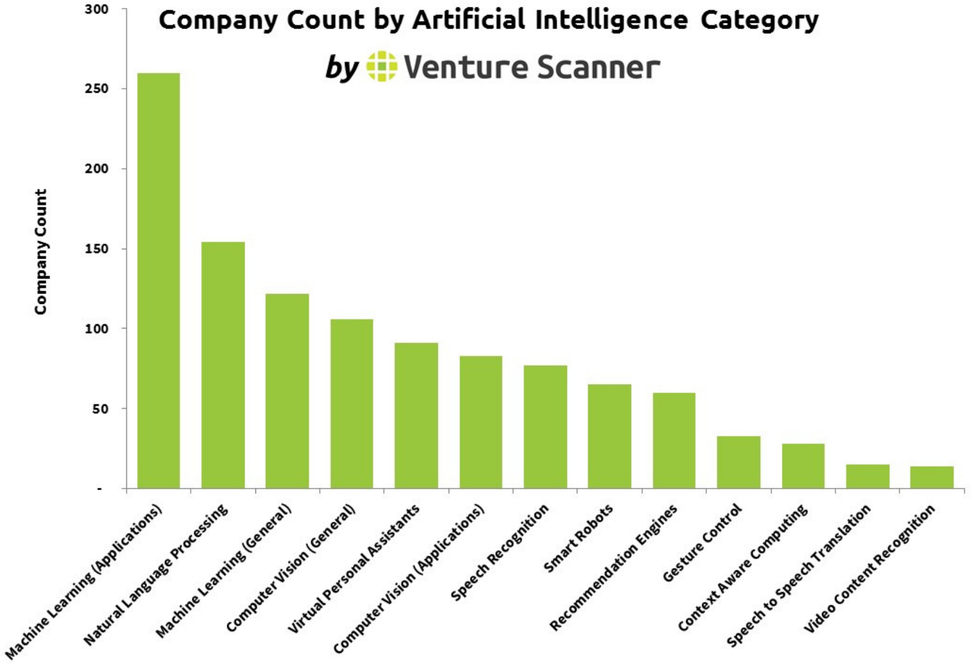


IA – ecossistema de empresas

REF: https://www.tractica.com/newsroom/press-releases / the-artificial-intelligence-market-ecosystem-is-Expanding-and-diversifying-rapidly-with-more-than-1000-companies-actively-driving-innovation-in-the-market/

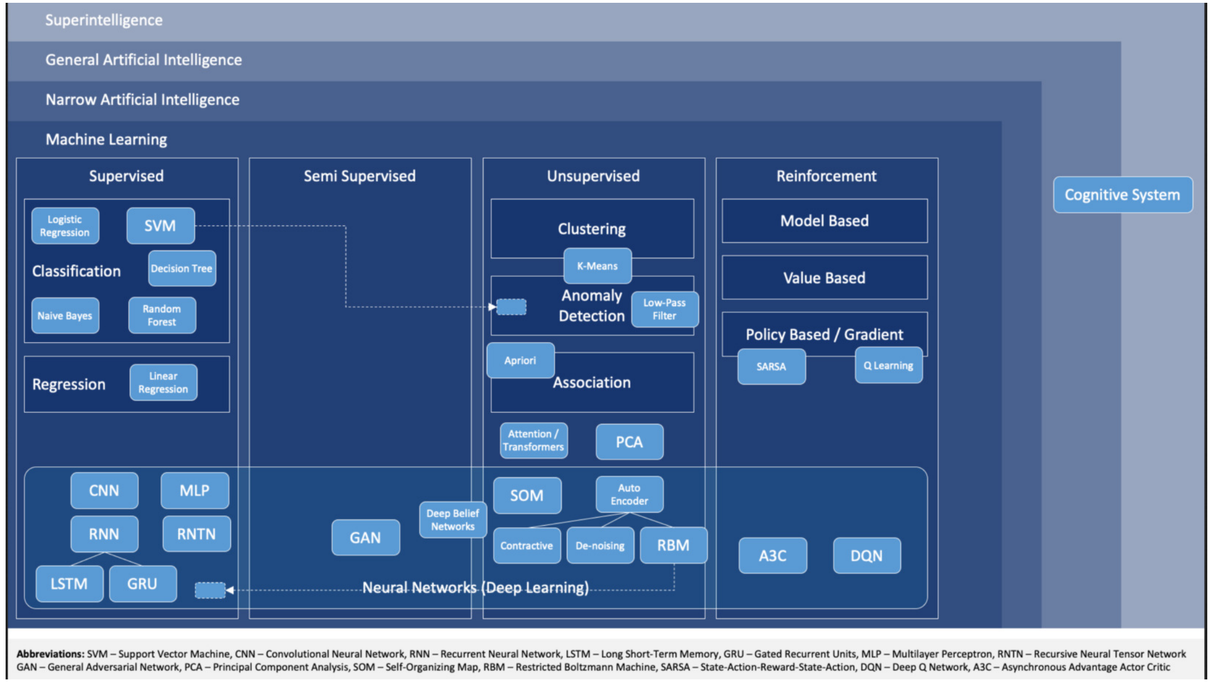


REF: https://www.venturescanner.com/tag/top-artificial-intelligence-companies/

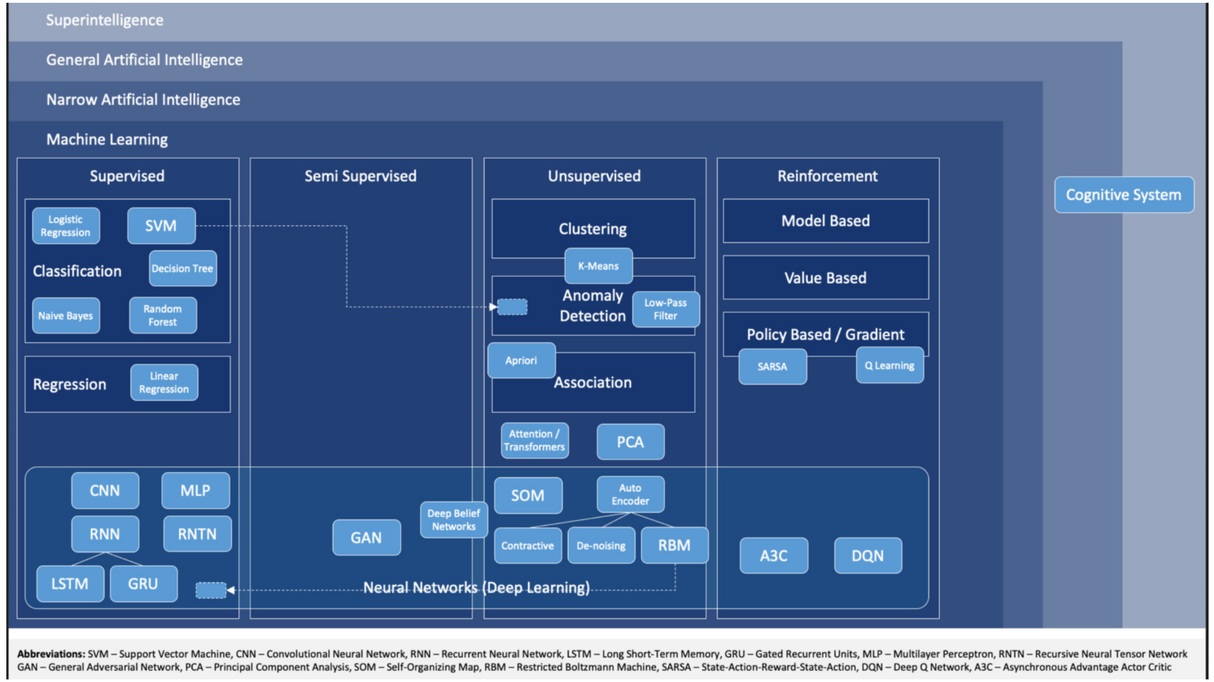


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8

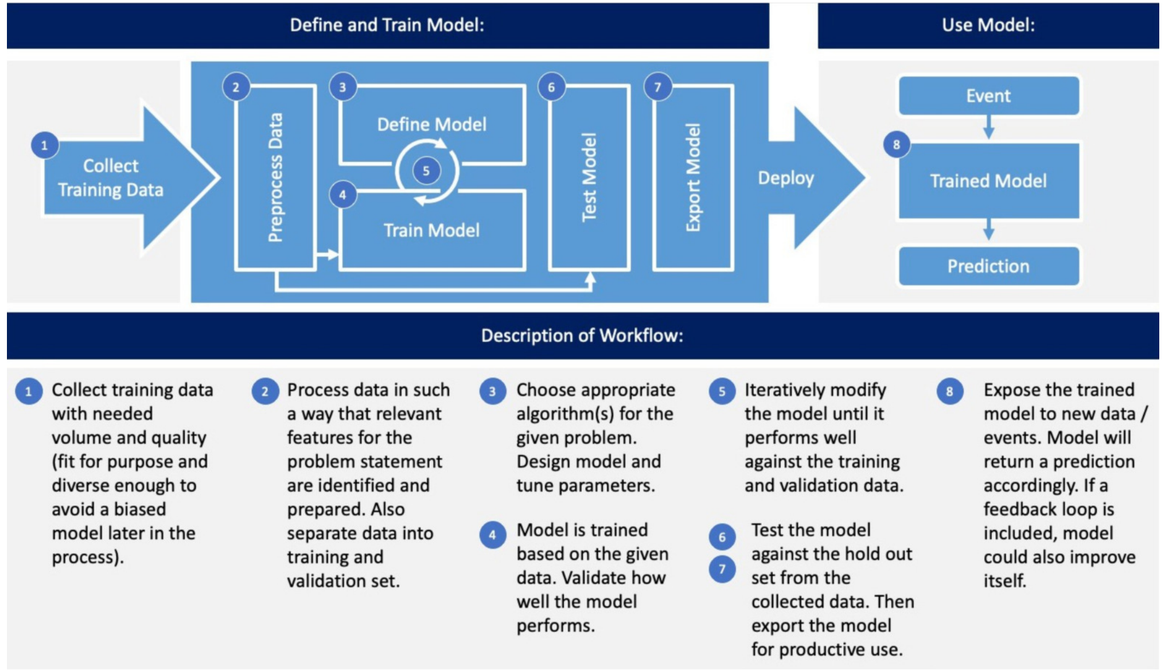


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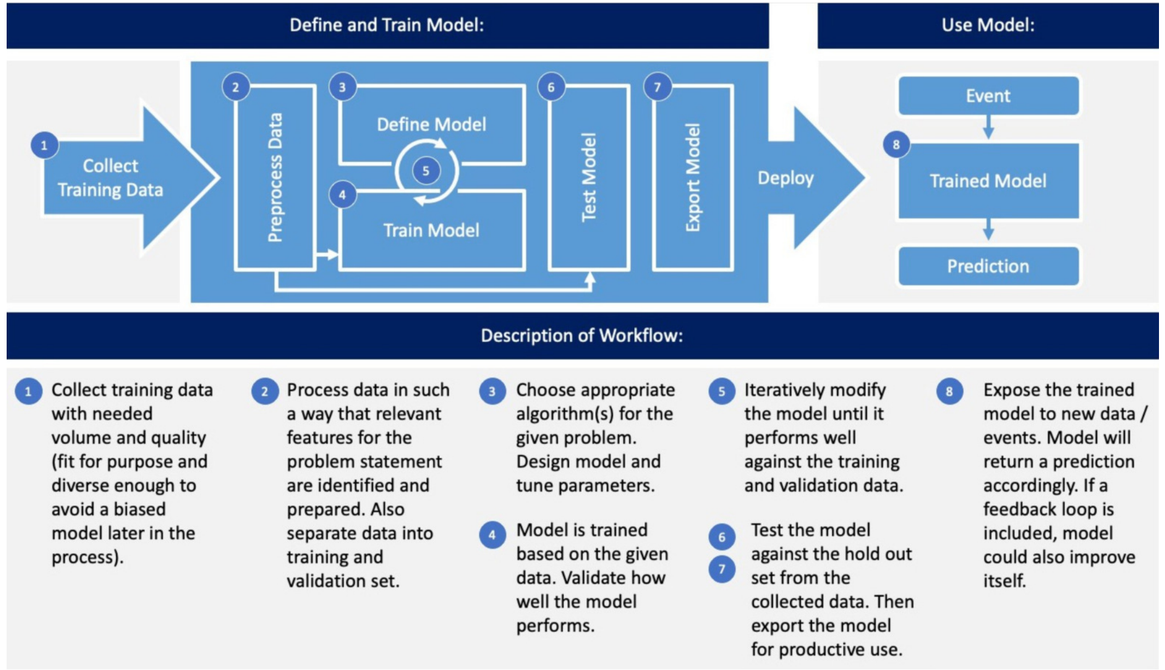
REF: https://towardsdatascience.com/artificial-intelligence-framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 -
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Workflow da APRENDIZAGEM DE MÁQUINA SUPERVISIONADA



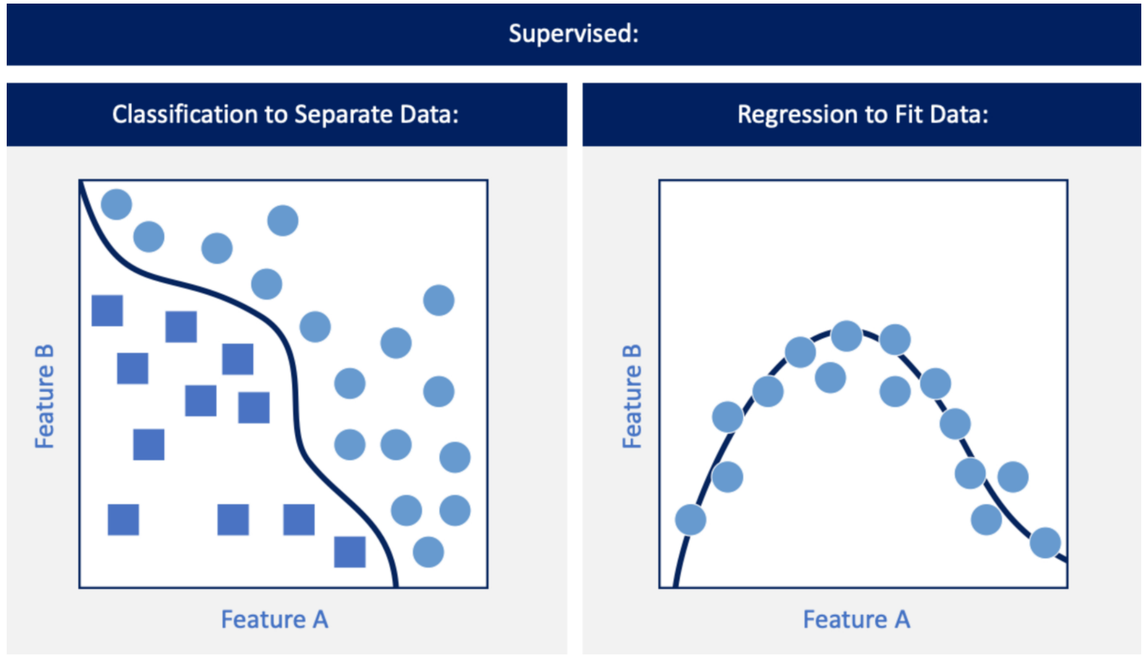
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Workflow da APRENDIZAGEM DE MÁQUINA SUPERVISIONADA



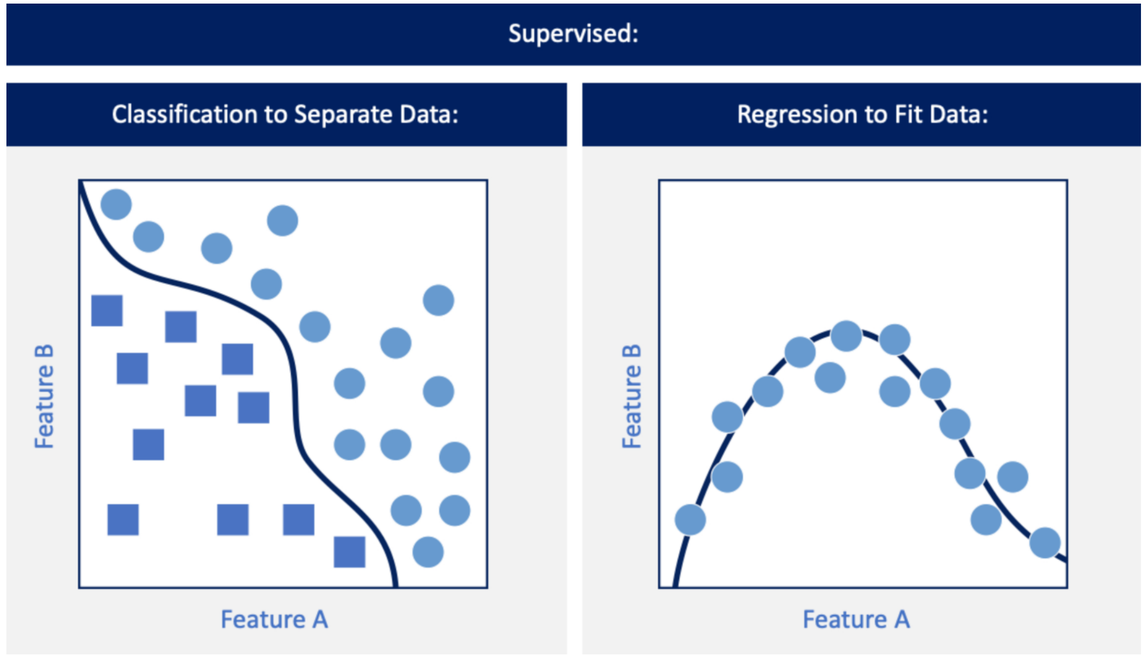
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APRENDIZAGEM DE MÁQUINA – SUPERVISIONADA TAREFAS



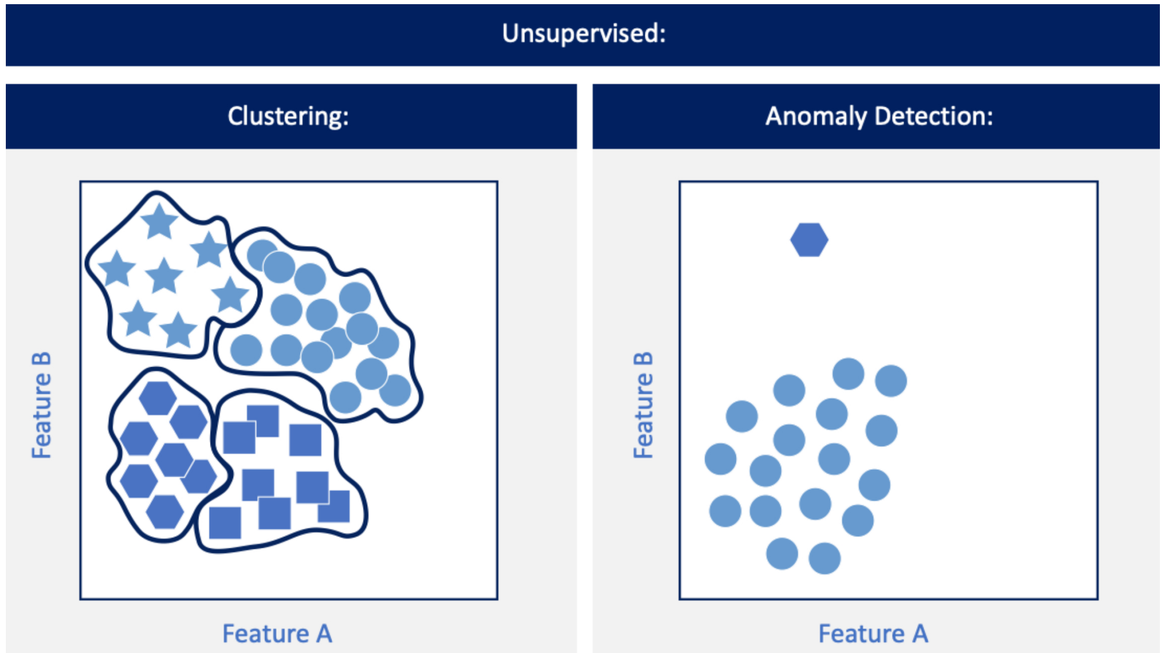
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APRENDIZAGEM DE MÁQUINA – TAREFAS SUPERVISIONADAS



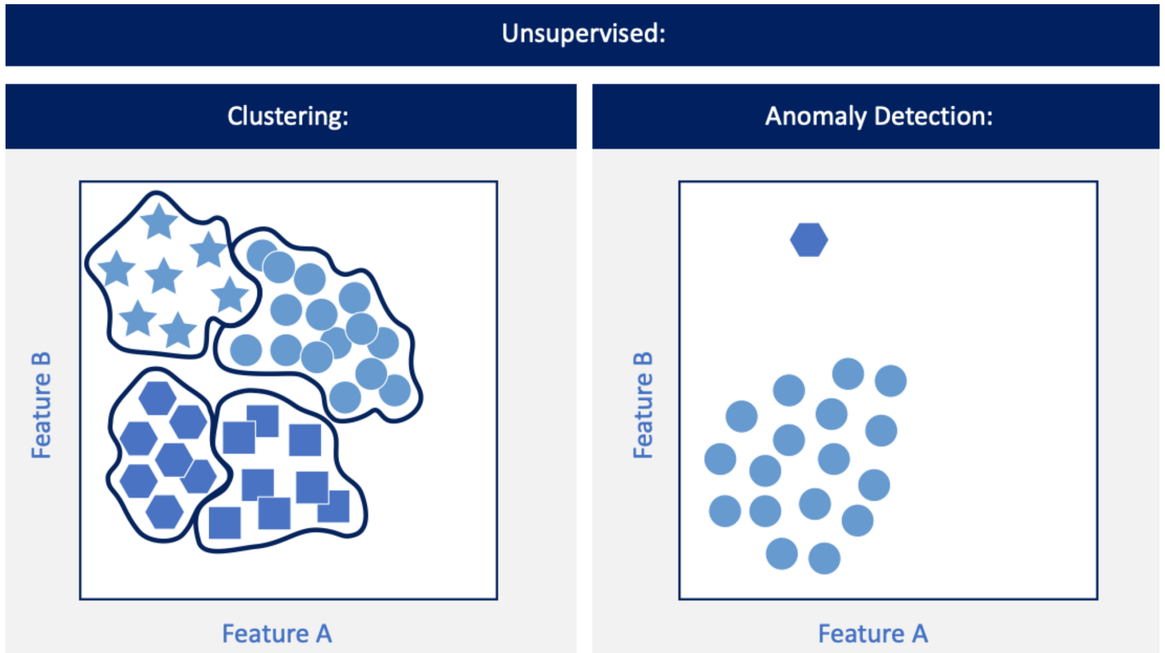
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APRENDIZAGEM DE MÁQUINA NÃO-SUPERVISIONADA
TAREFAS



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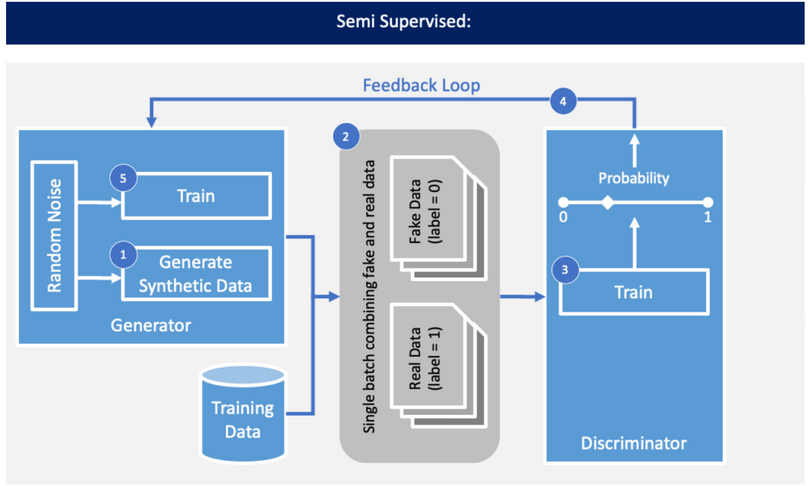
APRENDIZAGEM DE MÁQUINA NÃO-SUPERVISIONADA
TAREFAS



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Workflow da APRENDIZAGEM DE MÁQUINA SEMI-SUPERVISIONADA

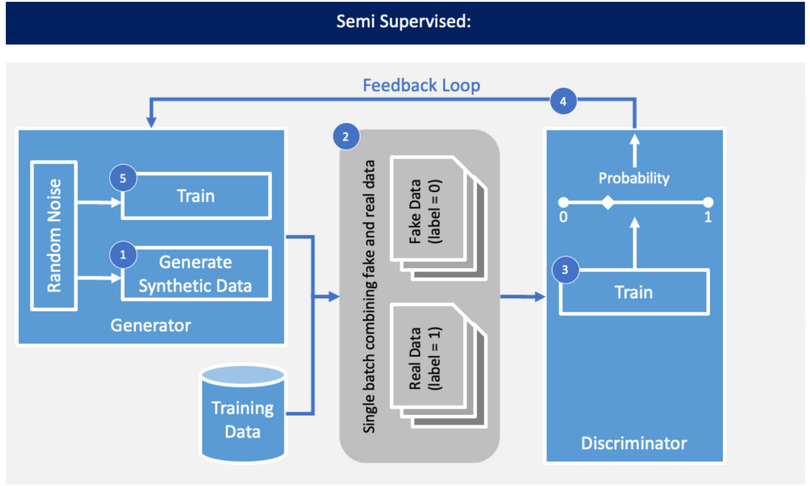


Neural Network - GAN (general adversarial network)

1. Synthetic data with the same dimension as the training data is generated based on a given noise vector.
2. Real training data and fake synthetic data is combined in a batch including the labels (fake or not).
3. Discriminator is trained with the goal to detect as many fake data points as possible and optimize its loss function.
4. Feedback from discriminator is „leaked“ for the generator in order to improve its opposing loss function.
5. Generator is trained with the goal to generate fake data that will not be classified as fake by discriminator.

REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

Workflow da APRENDIZAGEM DE MÁQUINA SEMI-SUPERVISIONADA



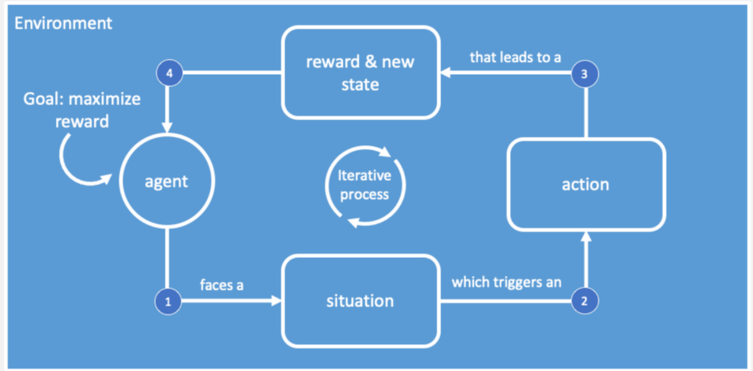
Rede Neural - GAN (rede adversária geral)

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APRENDIZAGEM DE MÁQUINA POR REFORÇO
(PRESSUPÕE INTERAÇÃO COM ALGUM MEIO- Princípio Cibernético)

Reinforcement Learning:



1. An agent faces a certain situation (known or unknown) within a given environment.
2. In this situation, the agent takes an action based on existing knowledge. With unknown situations, a random action is taken.
3. The outcome of the action will result in a new state / situation. Also, if the action is favorable towards the overall goal, the agent receives a reward.
4. Until the overarching goal is not achieved, the agent will continue with the process. With each iteration, positive actions are reinforced.

APRENDIZAGEM
por
REFORÇO

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APRENDIZAGEM DE MÁQUINA POR REFORÇO
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Reinforcement Learning:



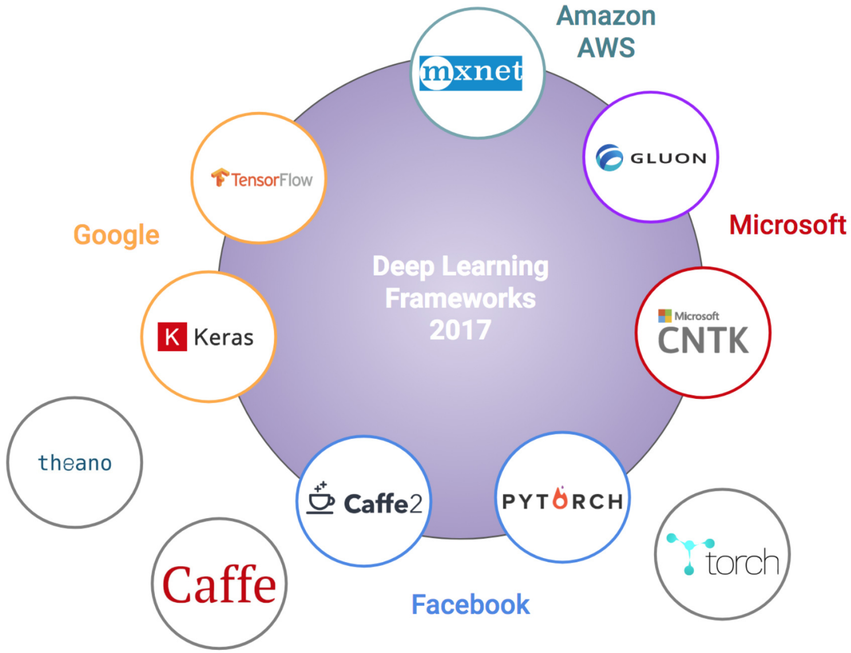
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APRENDIZAGEM por
REFORÇO

14

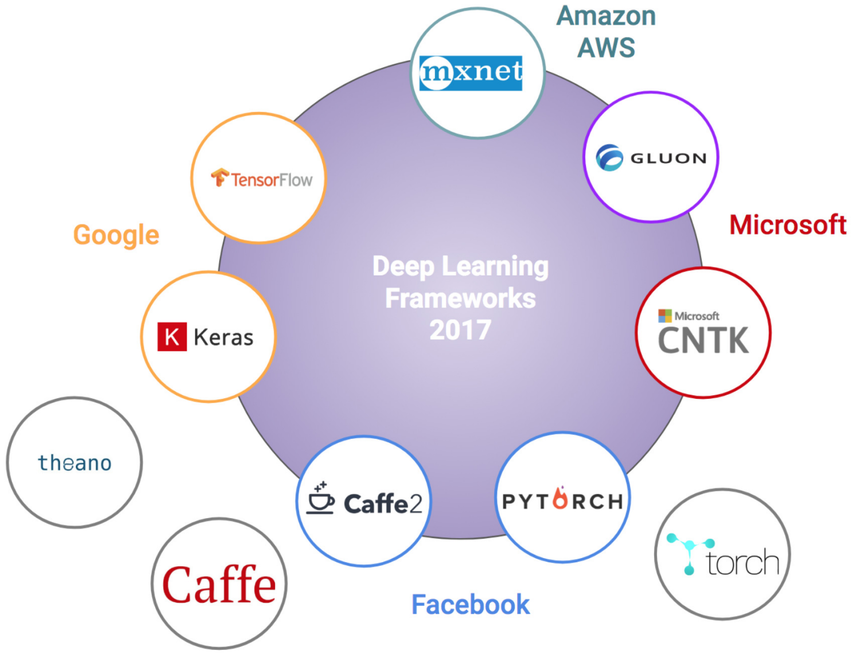
REF: https://towardsdatascience.com/artificial-intelligence-framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 -
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Como era em 2017: “Frameworks” de Deep Leaning



REF: https://towardsdatascience.com/battle-of-the-deep-learning-frameworks-part-i-cff0e3841750

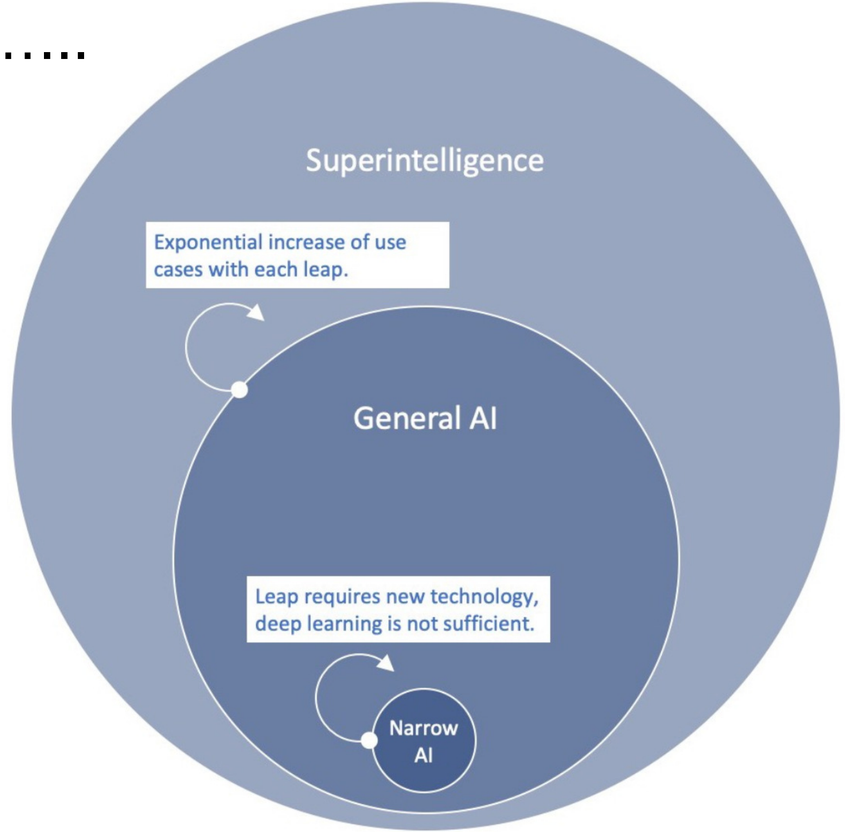
Como era em 2017: “Frameworks” de Deep Leaning



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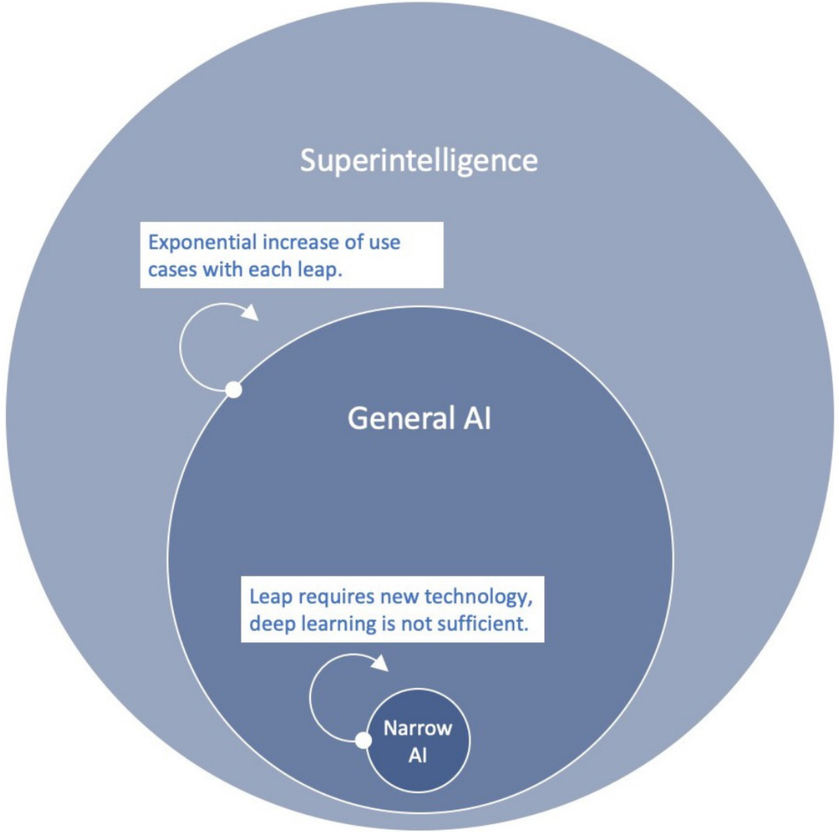
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MAS……



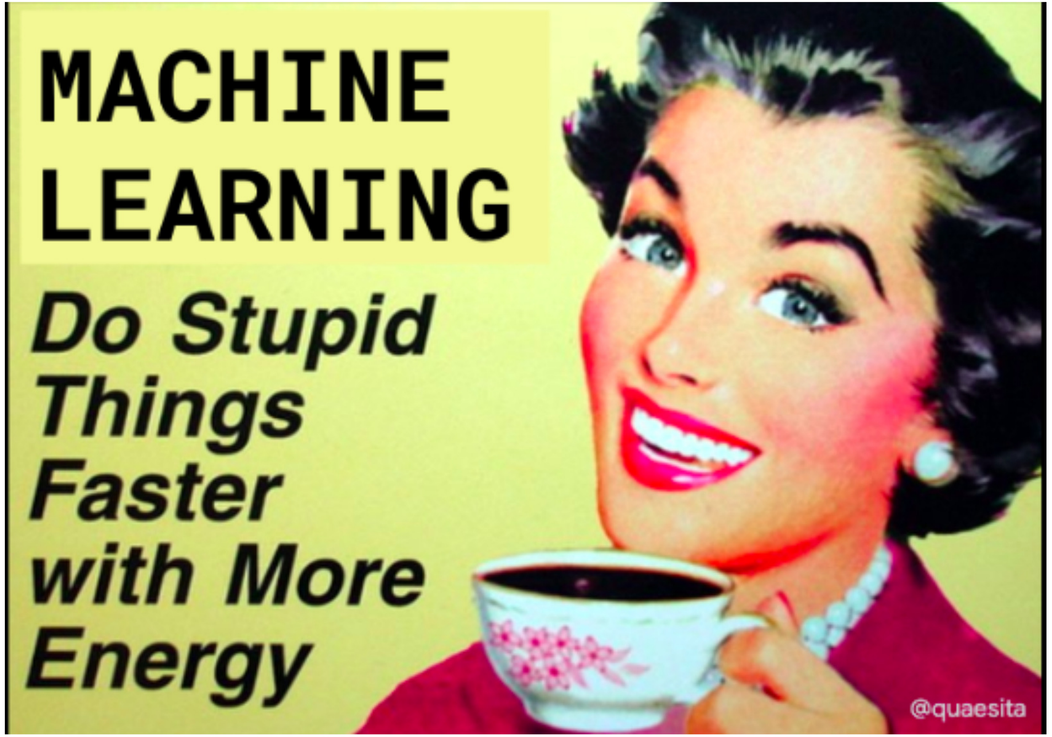
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MAS……



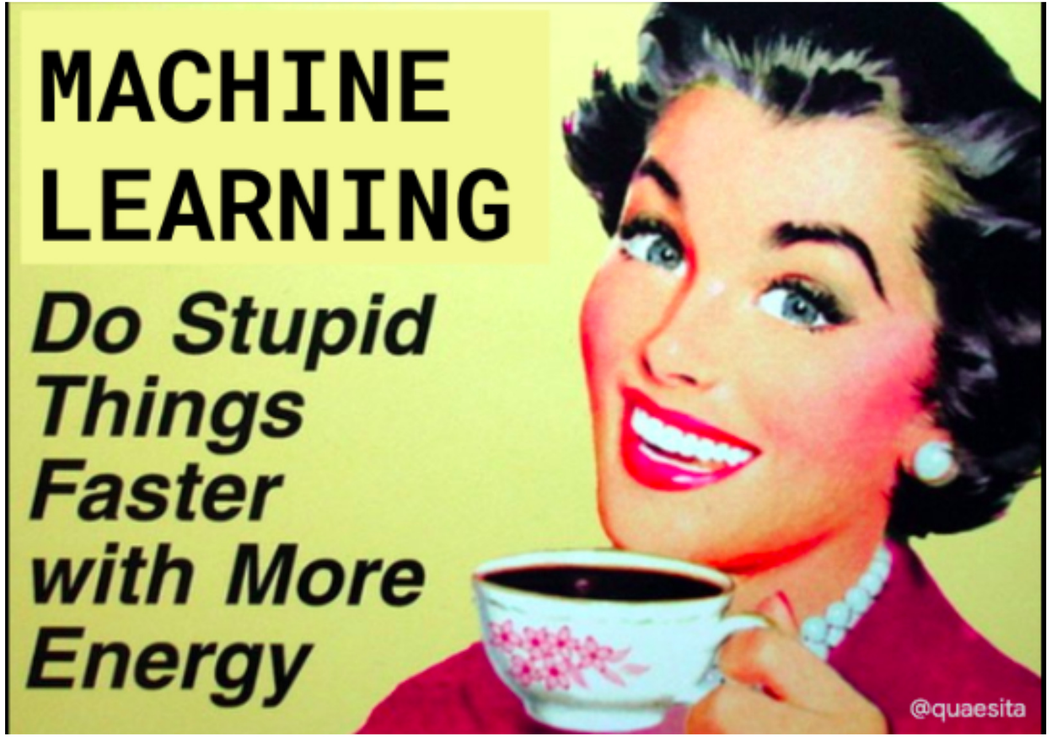
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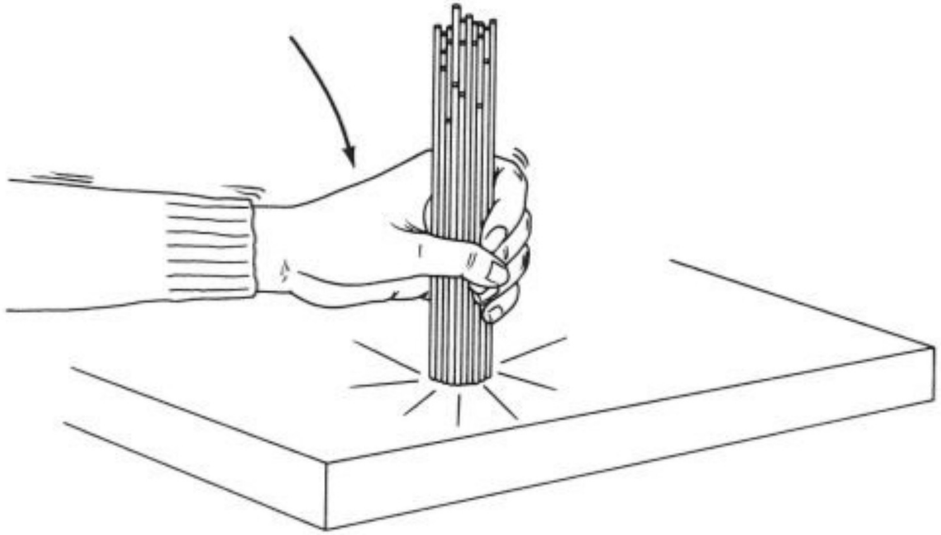
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APRENDIZAGEM DE MÁQUINA



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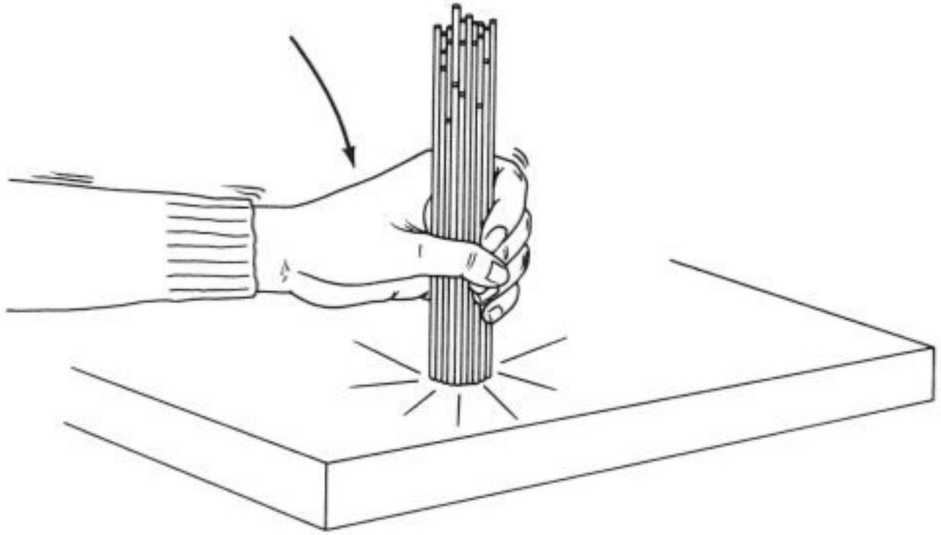
ORDENAÇÃO SPAGHETTI:
Computar dessa maneira requer CORPO FÍSICO (como a Computação DNA)



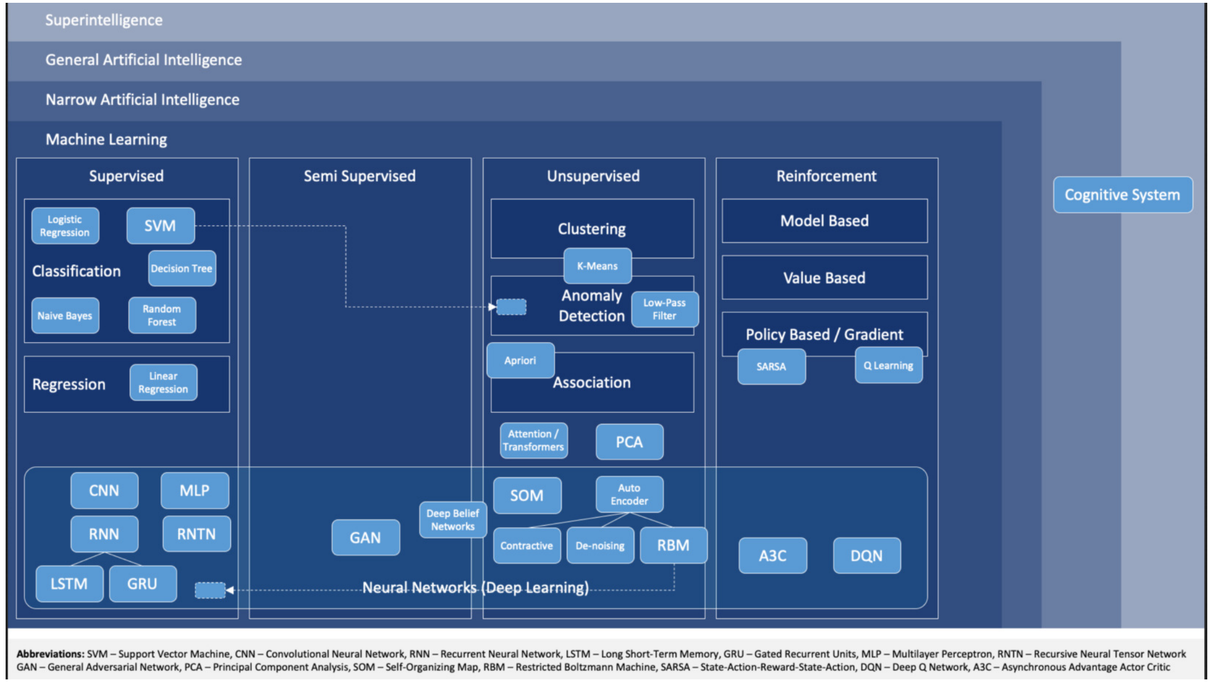
REF: https://towardsdatascience.com/the-nature-of-human-intelligence-from-a-computer-science-perspective-30f6e9c09146

LIMITES

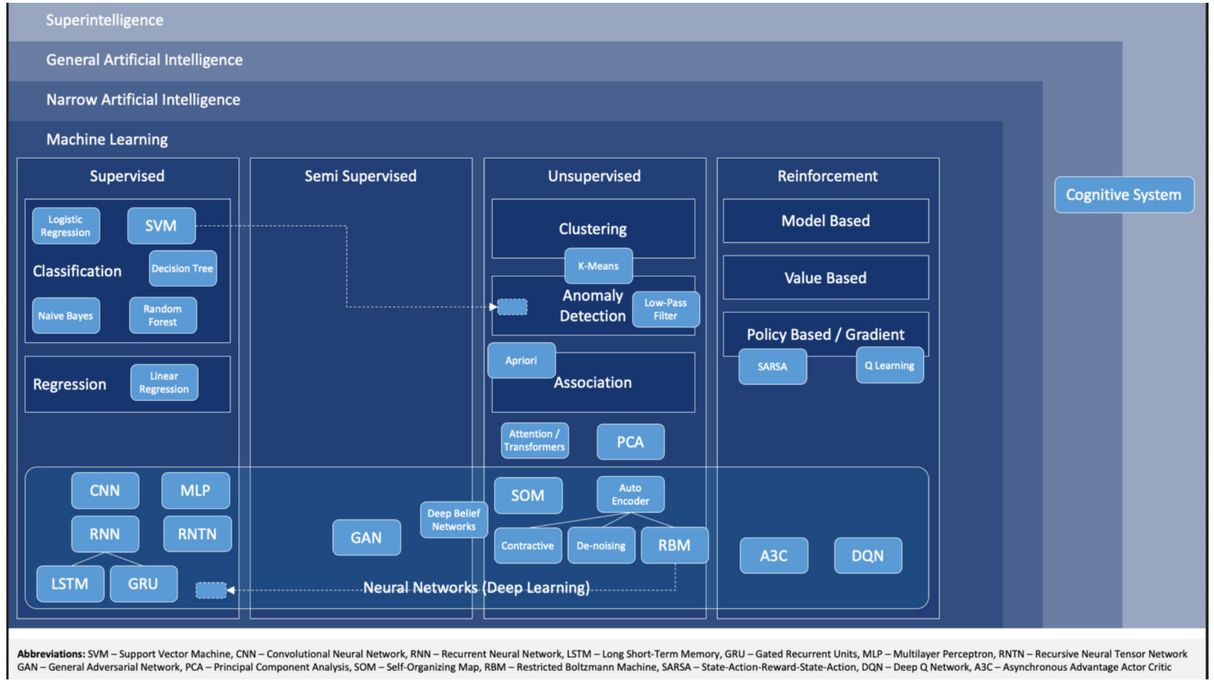
ORDENAÇÃO SPAGHETTI: Computar dessa maneira requer CORPO FÍSICO (como a Computação DNA)



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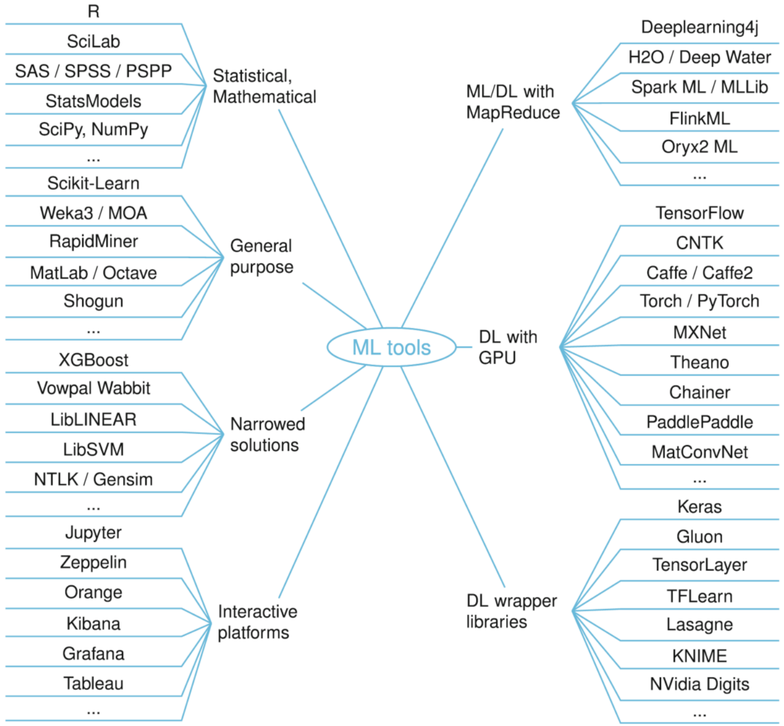


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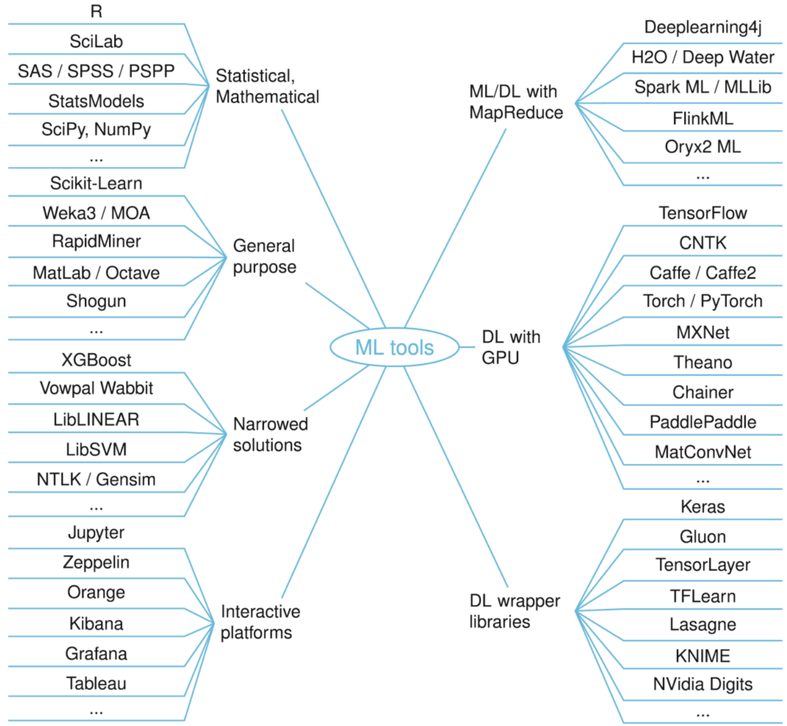
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ML-2019: Bibliotecas e Arcabouços (Frameworks)



REF:Artificial Intelligence Review (2019) 52:77–124
https://doi.org/10.1007/s10462-018-09679-

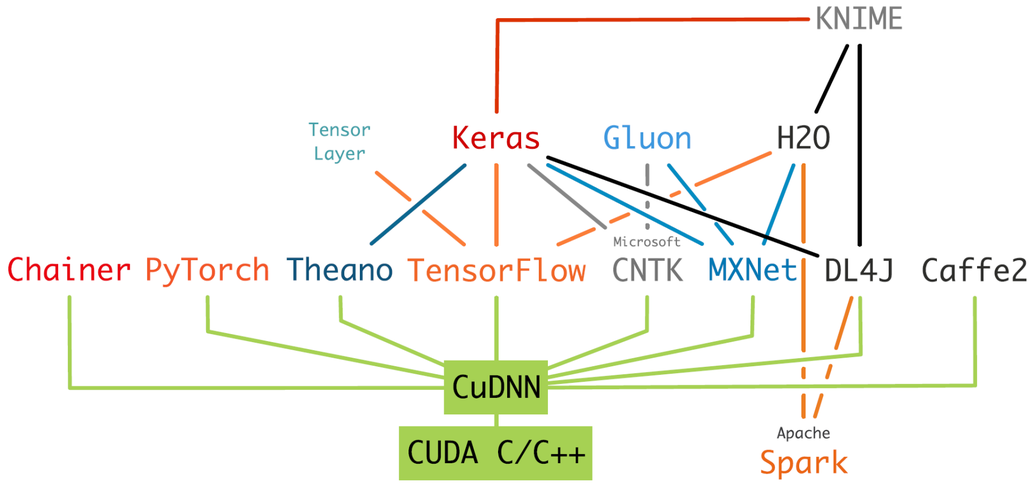
ML-2019: Bibliotecas e Arcabouços (Frameworks)



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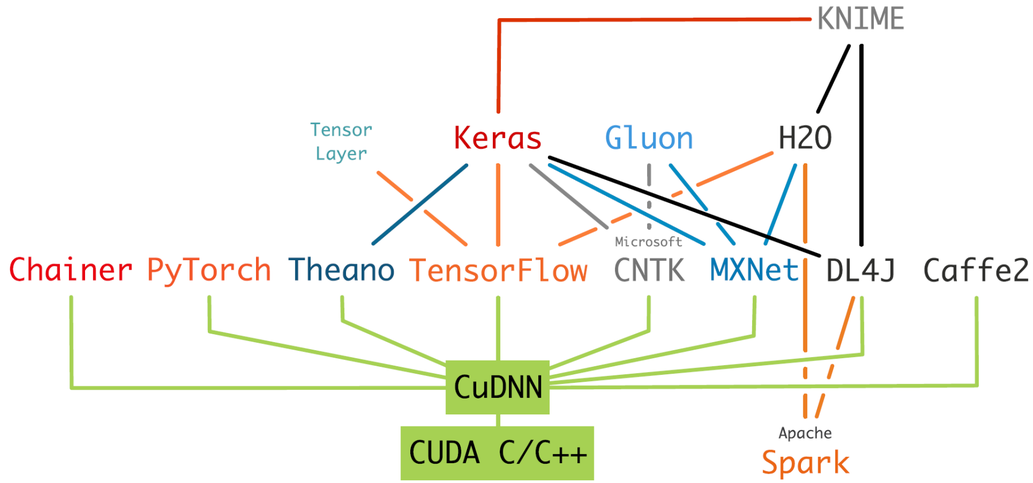
REF:Revisão de Inteligência Artificial (2019) 52:77–124
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Deep Learning -2019: Bibliotecas e Arcabouços (Frameworks) em níveis de abstração



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Tool	Licence	Written in	Computation graph	Interface	Popularity	Usage	Creator (notes)
TensorFlow (Numerical framework)	Open source, Apache 2.0	C++, Python	Static with small support for dynamic graph	Python, C++[a], Java[a], Go[a]	Very High Growing very fast	Academic Industrial	– **Google**
Keras (Library)	Open source, MIT	Python	Static	Python Wrapper for TensorFlow, CNTK, DL4J, MXNet, Theano	High Growing very fast	Academic Industrial	F. Chollet
CNTK (Framework)	Open source, Microsoft permissive license	C++	Static	Python, C++, BrainScript, ONNX	Medium Growing fast	Academic Industrial Limited mobile solution	– **Microsoft**
Caffe (Framework)	Open source, BSD 2-clause	C++	Static	C++, Python, MatLab	High Growing fast	Academic Industrial	Y. Jia **BAIR**
Caffe2 (Framework)	Open source, Apache 2.0	C++	Static	C++, Python, ONNX	Medium-low Growing fast	Academic Industrial Mobile solution	Y. Jia **Facebook**
Torch (Framework)	Open source, BSD	C++, Lua	Static	C, C++, LuaJIT, Lua, OpenCL	Medium–low Growing low	Academic Industrial	R. Collobert, K. Kavukcuoglu, C. Farabet
PyTorch (Library)	Open source, BSD	Python, C	Dynamic	Python, ONNX	Medium Growing very fast	Academic Industrial	A. Paszke, S. Gross, S. Chintala, G. Chanan
MXNet (Framework)	Open source, Apache 2.0	C++	Dynamic dependency scheduler	C++, Python, Julia, MatLab, Go, R, Scala, Perl, ONNX	Medium Growing fast	Academic Industrial	– **Apache**
Chainer (Framework)	Open source, Owners permissive license	Python	Dynamic	Python	Low Growing low	Academic Industrial	– **Preferred Networks**
Theano (Numerical framework)	Open source, BSD	Python	Static	Python	Medium-low Growing low	Academic Industrial	Y. Bengio **University of Montreal**

[a]Not fully covered

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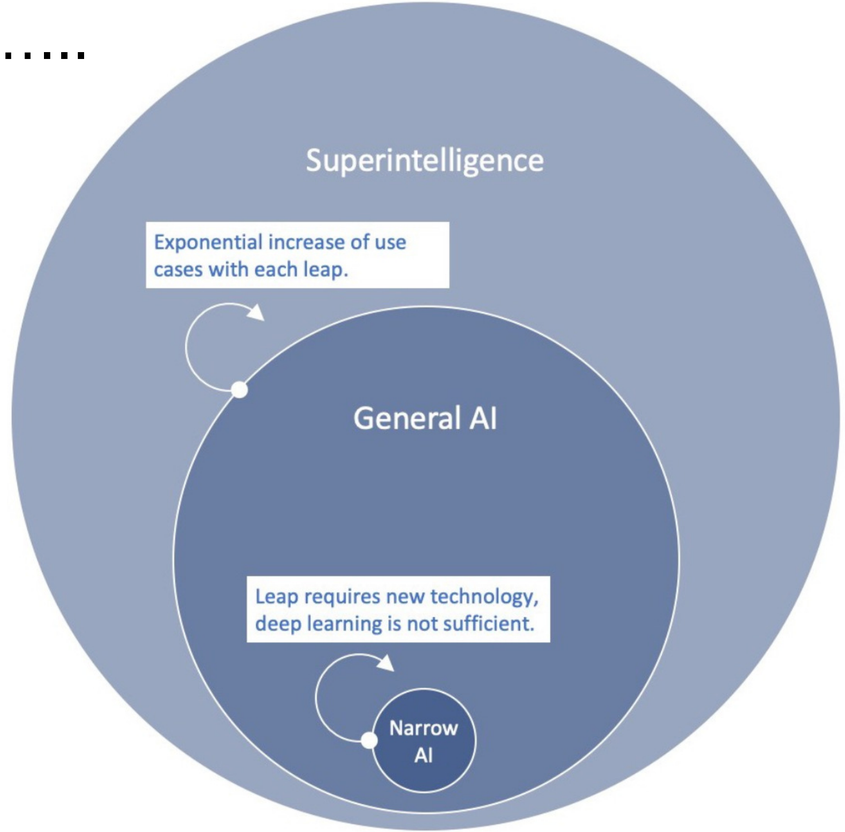
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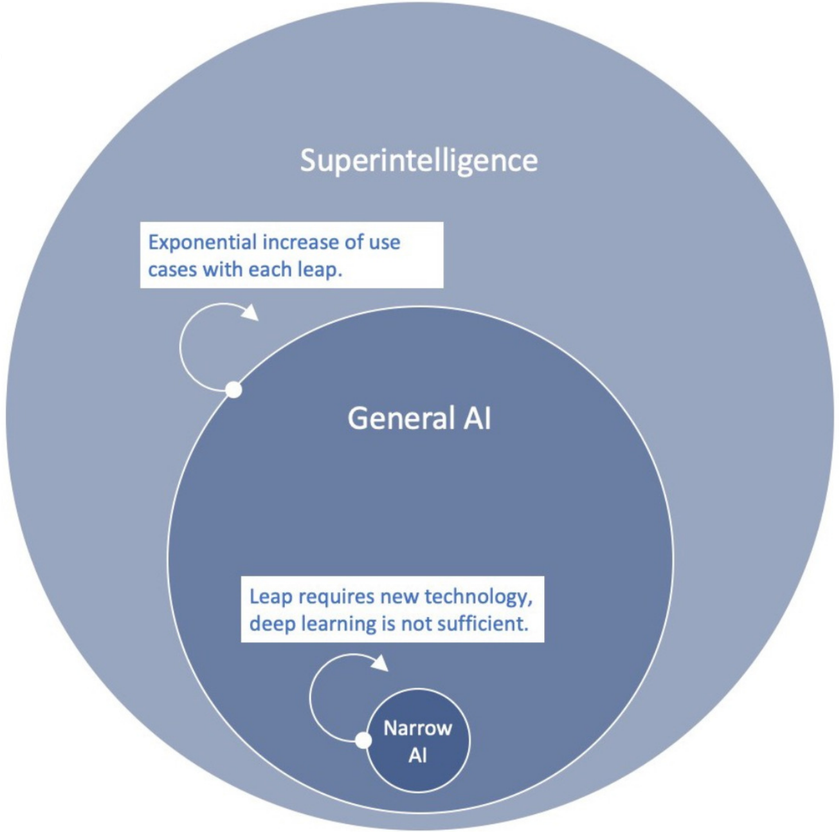
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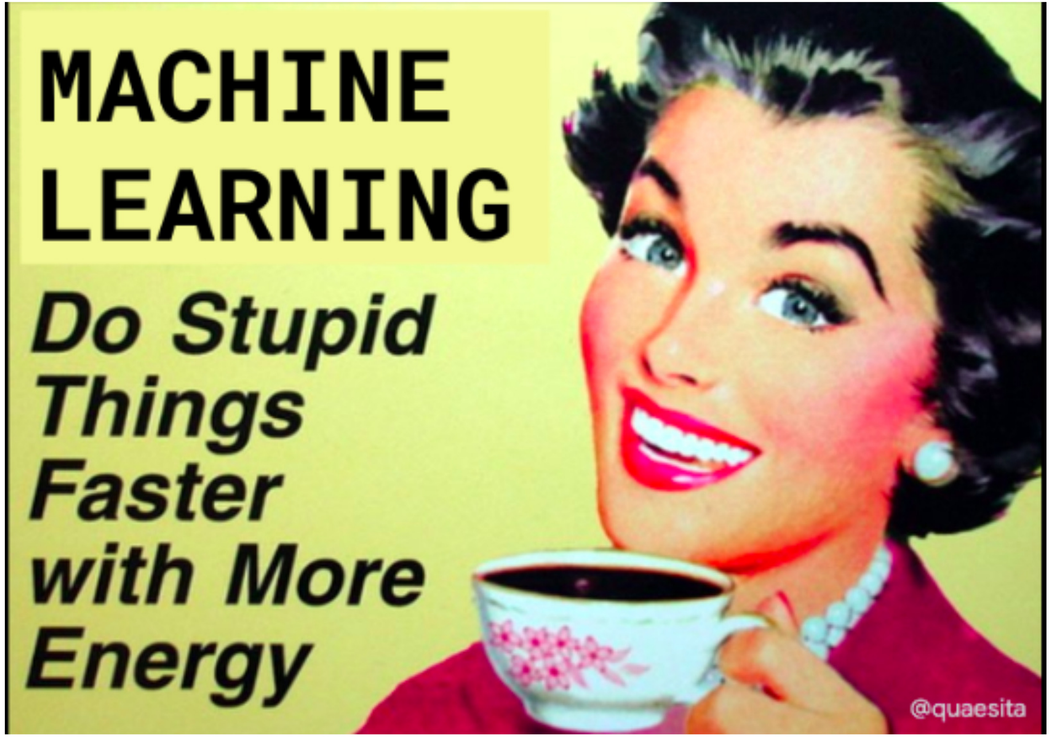
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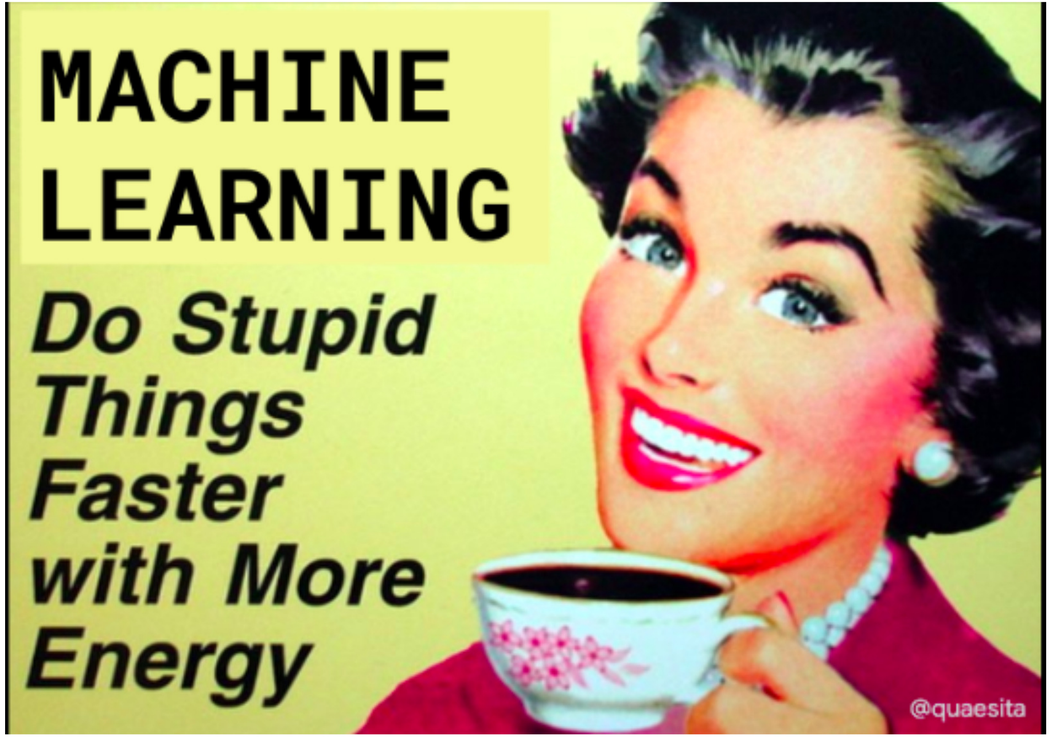
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APRENDIZAGEM DE MÁQUINA



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Precisamos ESTUDAR MAIS PROFUNDAMENTE para sair da arena movediza das “buzz words”

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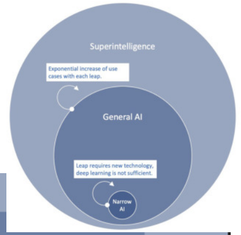
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25

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EXEMPLO: como funciona ?

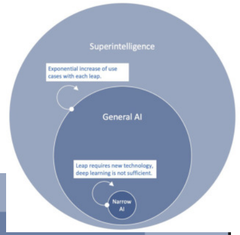


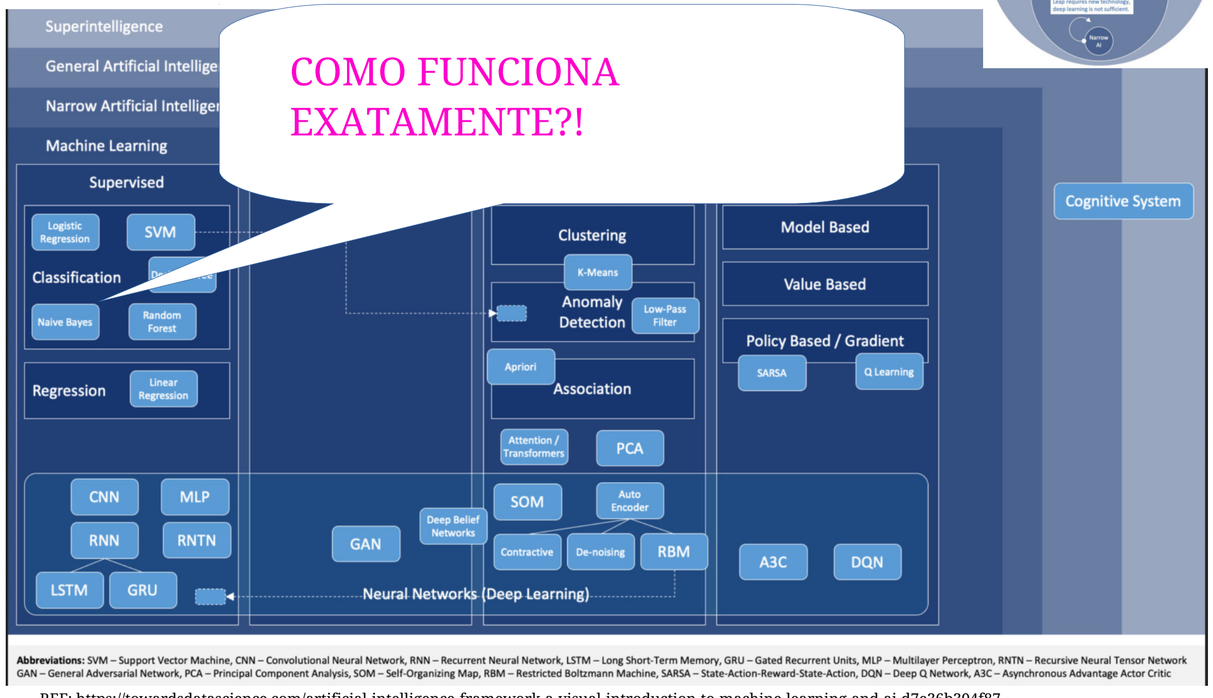


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XEMPLO: como funciona?

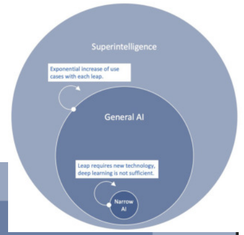




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EXEMPLO: como funciona ?



COMO FUNCIONA EXATAMENTE ?!

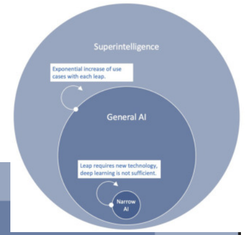


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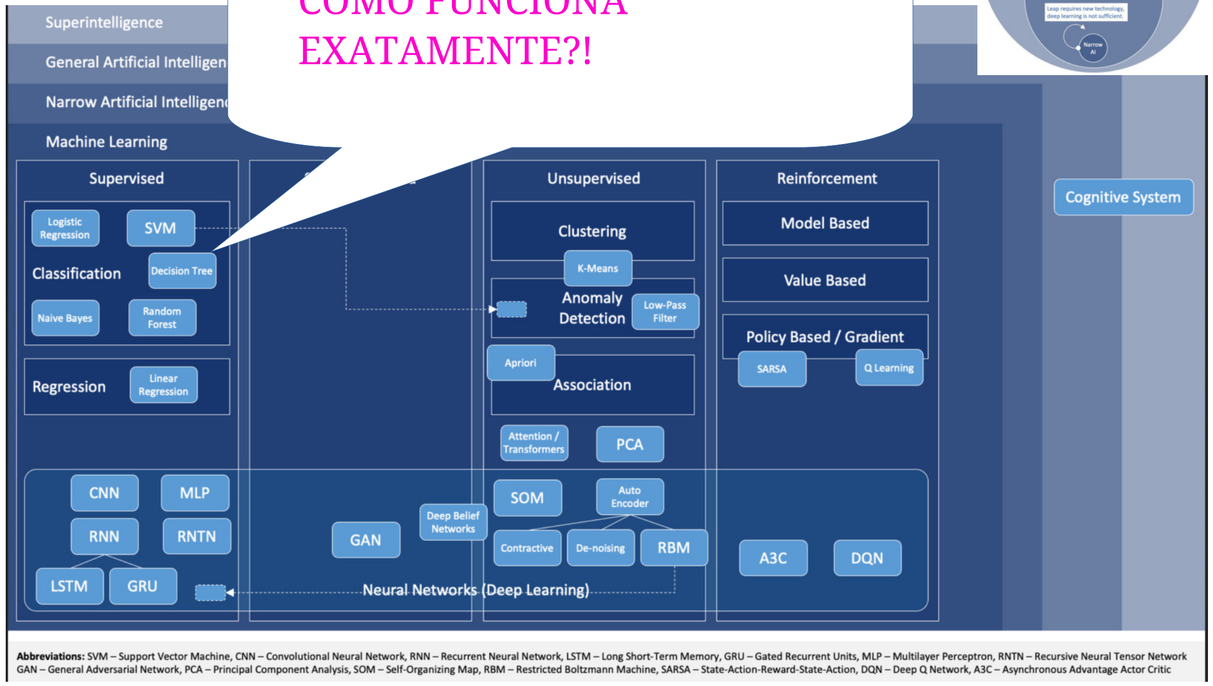
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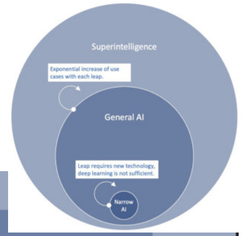


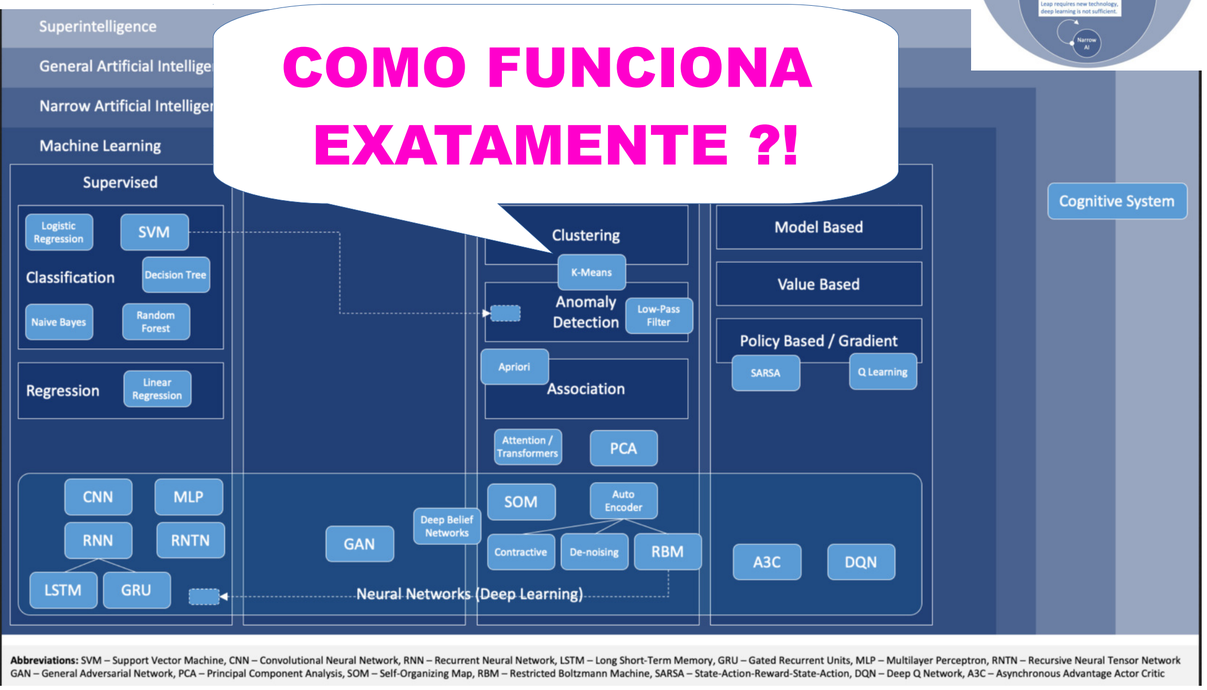
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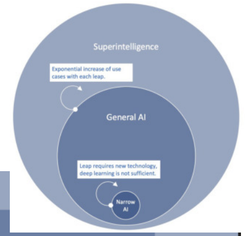


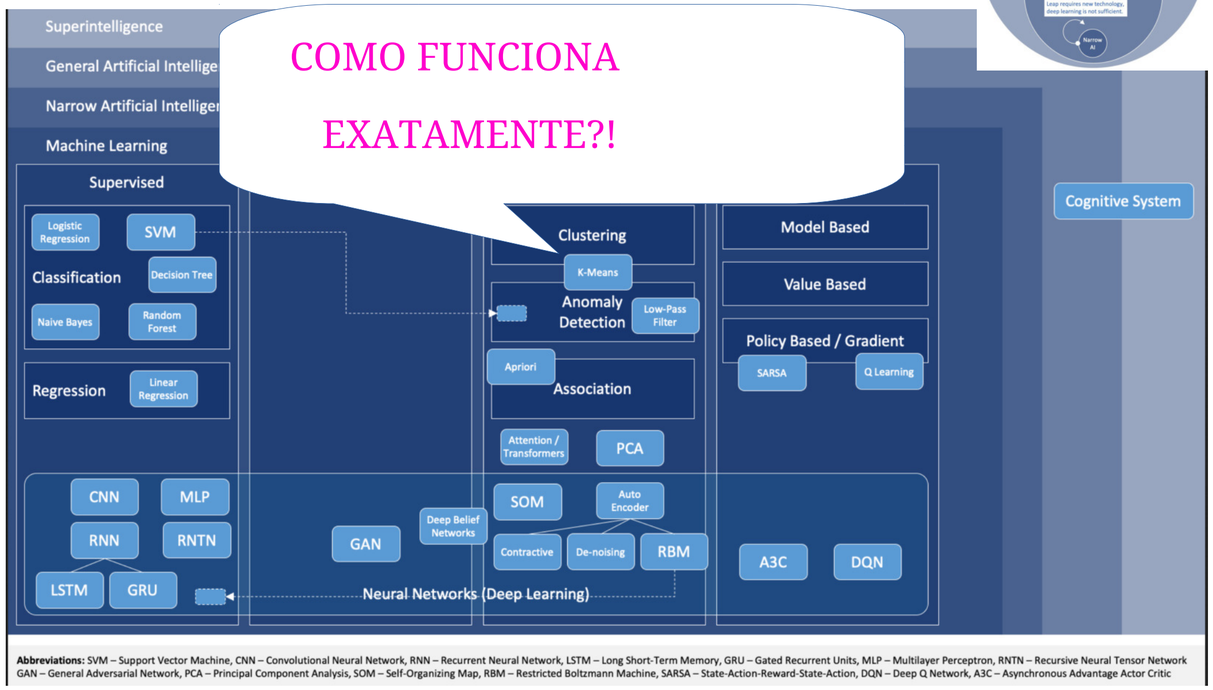


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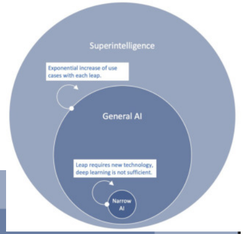




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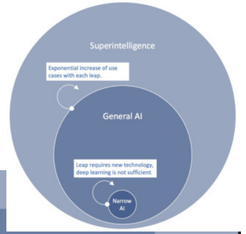


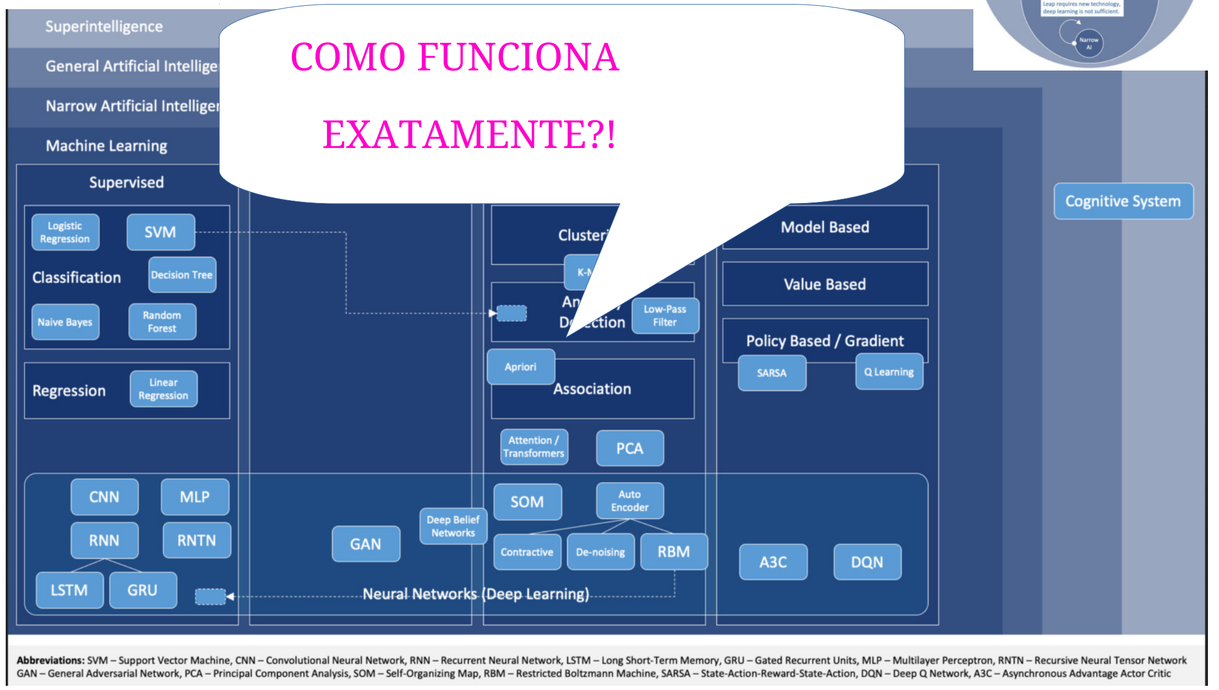


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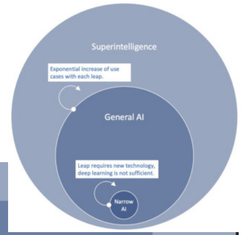




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O QUE É?
COMO FUNCIONA
EXATAMENTE ?!

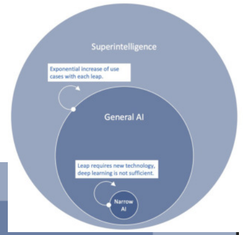


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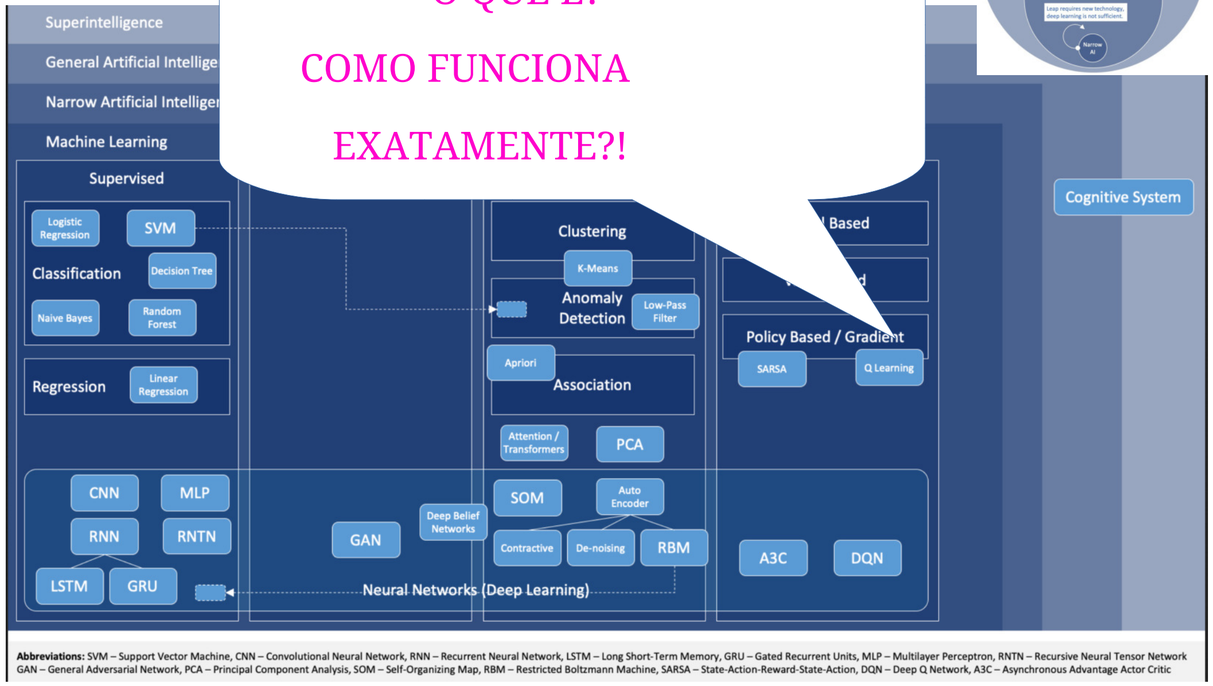
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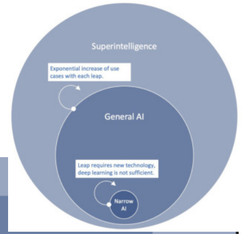


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COMO FUNCIONA

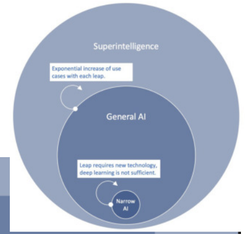
EXATAMENTE ?!



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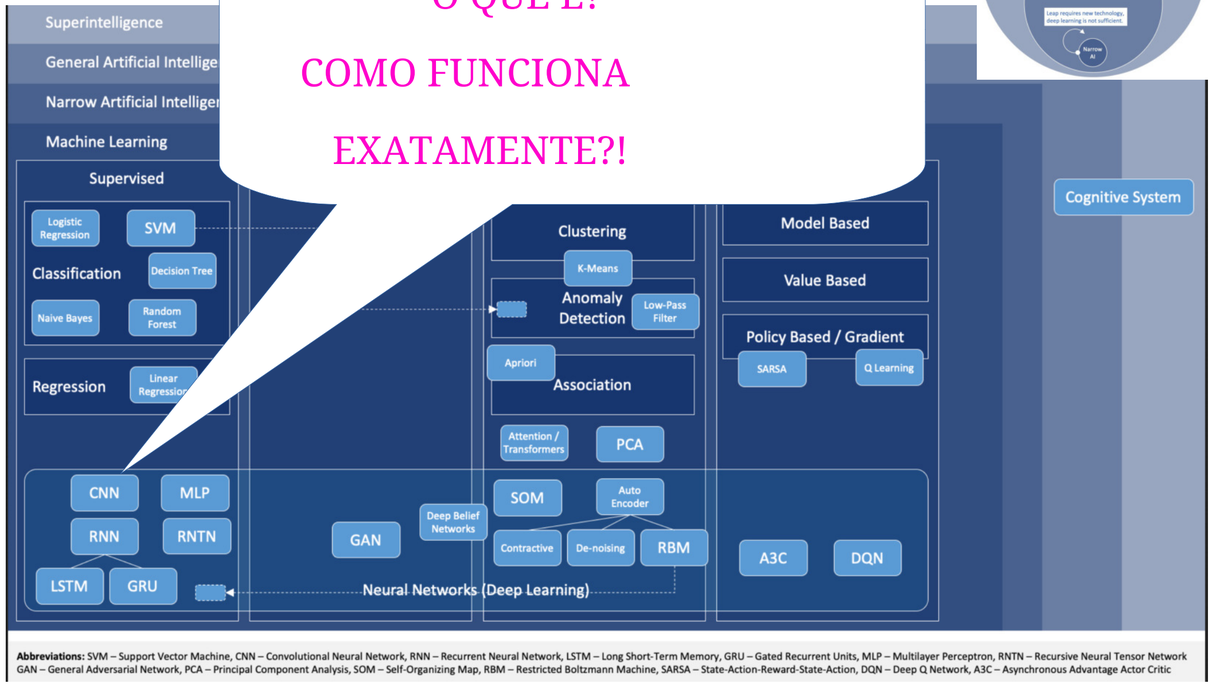
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